

Colony Growth and Development Model

A coupled mathematical model integrating:

- **WTH** = *World Tamer's Handbook* (TNE 0311) — colonial economic sectors and political mechanics
- **GURPS RM** = *GURPS Realm Management* — governance, revenue, and disruption framework
- **3turchin** = Turchin 2013 — structural-demographic political stress equations

Setting: SF colony on a habitable world ~1 year travel time from Earth (many parsecs). Colonists are functionally isolated between supply convoys. Time unit throughout: **months** (matching WTH's monthly turn cycle). Distances and masses in SI units; temperature in °C (human-scale); currency in colony credits [cr].

State Variables

Symbol	Meaning	Units
$N(t)$	total colony population	persons
$E(t)$	elite population (administrators, officers, technical specialists)	persons
$e(t) = E/N$	relative elite fraction	dimensionless
$AC(t)$	agricultural capital units deployed	capital-units
$IC(t)$	industrial capital units deployed	capital-units
$MC(t)$	materials capital units deployed	capital-units
$PC(t)$	power capacity installed	MW

Symbol	Meaning	Units
AL(t)	agricultural laborers assigned	persons
IL(t)	industrial laborers assigned	persons
ML(t)	materials laborers assigned	persons
D_lab(t)	labor demand (full-economy analog)	persons
Y(t)	cumulative colony debt to Earth	cr
CL(t)	Citizen Loyalty index	dimensionless $\in [0, 1]$
PT(t)	Political Track level	integer $\in \{-3, \dots, +3\}$
w(t)	relative welfare index	dimensionless
$\varepsilon(t)$	relative elite income	dimensionless
$\Psi(t)$	Political Stress Indicator	dimensionless

I — Population Dynamics

Total population ODE

$$\dot{N} = r_0 \cdot h(\text{Hab}) \cdot \sigma(\text{S_N}) \cdot N + I(t)$$

Symbol	Meaning	Units
r_0	base annual growth rate = $0.02 \text{ y}^{-1} = 1/600 \text{ mo}^{-1}$	mo^{-1}
$h(\text{Hab})$	GURPS RM habitability modifier (see §V)	dimensionless
$\sigma(\text{S_N})$	nutrition stress modifier (see §III)	dimensionless
$I(t)$	immigration inflow from Earth	persons/month

Immigration

Colonist waves are requested at time t_{req} and arrive after a one-way transit of $\Delta = 12$ months:

$$I(t) = \sum_k I_k \cdot \delta(t - t_{\{k, \text{req}\}} - 12)$$

No communication faster than transit exists; a wave, once dispatched, cannot be recalled. The colony administrator's only levers before the wave lands are requests sent 12+ months earlier.

Population compartments

$$N_w = \lambda \cdot N \text{ (working-age population, } \lambda \approx 0.55)$$

$$N_c = N_w - E \text{ (common workers available for sector assignment)}$$

$$AFL = AL + IL + ML \text{ (assigned full laborers; } AFL \leq N_c)$$

II — Economic Sector Model

Effective production factor M (WTH formula, all sectors)

Given capital C_x and assigned labor L_x for sector x :

$$M_x(t) = L_x(t) \text{ if } L_x < C_x \text{ (labor-limited)}$$

$$M_x(t) = \min(C_x + 0.5 \cdot (L_x - C_x), 1.5 \cdot C_x) \text{ if } L_x \geq C_x \text{ (capital-limited with surplus-labor bonus)}$$

This encodes WTH's "output multiplier = $AC + 0.5 \times [AL - AC]$ " rule. Extra labor beyond capital yields half-rate marginal output, capped at $1.5 \times$ the capital level.

Power constraint

$$P_{\text{req}}(t) = p_A \cdot AC(t) + p_I \cdot IC(t) + p_M \cdot MC(t)$$

If $P_{avail}(t) < P_{req}(t)$, production in all underpowered sectors is penalized by factor P_{avail} / P_{req} . (Power deficit is a hard constraint; the colony must install capacity before capital can operate at full rate.)

Agricultural output

$$Q_A(t) = q_A \cdot M_A(t) \cdot R_A \cdot \phi(t) \cdot \eta(t) \cdot \min(1, P_{avail}/P_{req})$$

Symbol	Meaning	Units
Q_A	rations produced	rations/month
q_A	base rations per effective capital unit per month	rations/capital-unit/month
R_A	planetary richness rating (from survey, 1–5)	dimensionless
$\phi(t)$	growing-season modifier $\in [0.5, 1.0]$ (annual cycle)	dimensionless
$\eta(t)$	output roll multiplier $\sim \text{Uniform}(0.80, 1.20)$ (WTH d20 roll)	dimensionless

Industrial output

$$Q_I(t) = q_I \cdot M_I(t) \cdot \eta(t) \cdot \min(1, P_{avail}/P_{req})$$

Symbol	Meaning	Units
Q_I	credits produced	cr/month
q_I	base credits per effective capital unit per month	cr/capital-unit/month

Industrial output is the colony's monetary economy. It funds capital purchases, imports, debt service, and government operations.

Materials output

$$Q_M(t) = q_M \cdot M_M(t) \cdot \eta(t) \cdot \min(1, P_avail/P_req)$$

Symbol	Meaning	Units
Q_M	raw materials extracted	tonnes/month
q_M	base tonnes per effective capital unit per month	t/capital-unit/month

Materials feed capital construction (internal) or export (external).

GDP equivalent

$$G(t) = 12 \cdot [p_A \cdot Q_A(t) + Q_I(t) + p_M \cdot Q_M(t)]$$

Units: cr/year. p_A [cr/ration] and p_M [cr/tonne] are shadow prices converting physical outputs to a common credit value. $G(t)$ serves as the colony's GDP analog, driving all Turchin equations that require G .

III — Satisfaction Indices and Relative Wage

Standard of Nutrition

$$S_N(t) = Q_A(t) / (N(t) \cdot r_min)$$

Symbol	Meaning	Units
r_min	minimum adequate rations per person per month	rations/person/month

$S_N = 1 \rightarrow$ adequate. $S_N < 1 \rightarrow$ shortfall (starvation risk). $S_N > 1 \rightarrow$ surplus available for export.

Nutrition stress modifier feeding population growth:

$$\sigma(S_N) = \min(S_N, 1)^{\kappa_N} \text{ with } \kappa_N > 0$$

When $S_N \geq 1$, $\sigma = 1$ (no penalty). When $S_N < 1$, growth rate suppressed and mortality rises.

Standard of Shelter

| $S_S(t) = H(t) / N(t)$

where $H(t)$ = housing capacity [berths], grows through construction capital deployment.

$S_S = 1 \rightarrow$ every colonist housed. $S_S < 1 \rightarrow$ overcrowding.

Standard of Living

| $S_L(t) = Q_I(t) / (N(t) \cdot I_{\min})$

Symbol	Meaning	Units
$I_{\min}$	minimum adequate credits per person per month (goods, services)	cr/person/month

Composite relative welfare (Turchin w-analog)

| $w(t) = [S_N(t) + S_S(t) + S_L(t)] / 3$

This maps WTH's three satisfaction indices to the dimensionless relative wage w used throughout Turchin's structural-demographic equations. When all three standards sit at adequacy, $w = 1 = w_0$ (the neutral mobility threshold).

IV — Labor Market and Wages

Labor demand (monthly operational)

| $D_{\text{lab}}(t) = G(t) / (2000 \cdot 12 \cdot P_{\text{prod}}(t))$

Adapting Turchin's contemporary-US proxy (Eq. 14 source variable), where P_{prod} [cr/hour] is colony labor productivity (grows with Education Rating ER: $P_{\text{prod}} = P_0 \cdot (1 + \text{ER}/6)$).

Labor supply: $S_{\text{lab}}(t) = \lambda \cdot N(t)$

Wage (Turchin Eq. 9 / 10 adapted)

$W(t+\tau) = \acute{a} \cdot (G(t)/N(t))^{\alpha} \cdot (D_{\text{lab}}(t)/S_{\text{lab}}(t))^{\beta} \cdot C(t)^{\gamma}$

$w(t+\tau) = W(t+\tau) / (G(t)/N(t))$

Symbol	Meaning	Units
τ	wage-stickiness lag (contract cycles)	months
$\acute{a}$	scale parameter	dimensionless
α	GDP-per-capita elasticity	dimensionless (≈ 1)
β	labor-market elasticity	dimensionless (≈ 0.5)
$C(t)$	colonial authority coercive factor (charter rules, CR)	dimensionless
γ	coercion elasticity	dimensionless

Note: In a colonial setting $C(t)$ is non-trivial — the charter company can suppress wages through contract labor laws. $C(t)$ is indexed to the colony's Control Rating (GURPS RM CR): higher CR $\rightarrow$ lower $C \rightarrow$ wage suppression.

The w computed here should converge with the satisfaction-based $w(t)$ from §III; in practice use the satisfaction index as the observable proxy and this equation as the predictive mechanism.

V — Habitability and Planetary Parameters (GURPS RM)

Habitability modifier

GURPS RM rates habitability 0 (Disastrous) to 19 (Excellent). Define:

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$$h(\text{Hab}) = \text{Hab} / 10 \text{ (normalized, } 0 \rightarrow 0, \approx 10 \rightarrow 1, 19 \rightarrow 1.9)$$

Habitability affects:

- Population growth: via $h(\text{Hab})$ in §I
- Agricultural carrying capacity: $K_A = K_0 \cdot h(\text{Hab})$ (more habitable $\rightarrow$ more arable land)
- Agricultural richness: R_A partially determined by Hab

Education Rating modifier on labor productivity

GURPS RM Education Rating $ER \in [0, 6]$:

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$$P_{\text{prod}}(t) = P_0 \cdot (1 + ER(t)/6)$$

ER grows slowly as the colony builds schools; it determines the productivity of each labor-unit and thus real labor demand.

Control Rating and the coercive wage factor

CR	C(t)	Interpretation
2	1.3	Free society; wages set by market
3	1.1	Mild regulation
4	1.0	Standard colonial charter
5	0.85	Heavy state control
6	0.70	Near-mandatory labor

VI — Elite Dynamics (Turchin Eqs. 4 / 13)

Elite population ODE

$$\dot{E}(t) = r_e \cdot E(t) + \mu_0 \cdot (w_0 - w(t)) / w(t) \cdot N(t)$$

Simplified (assuming $r_e \approx r_0$ so $e = E/N$ evolves independently of N):

$$\dot{e}(t) = \mu_0 \cdot (w_0 - w(t)) / w(t)$$

Symbol	Meaning	Units
r_e	elite natural growth rate $\approx r_0$	mo^{-1}
μ_0	social mobility rate parameter	mo^{-1}
w_0	mobility-neutral welfare threshold = 1	dimensionless

When $w < 1$ (depressed welfare), upward mobility accelerates → elite numbers swell. This is the core instability mechanism: poor colony conditions produce more people seeking elite positions than the administrative structure can absorb.

Colony-specific modification: elite positions are fixed by the administrative org-chart. Available slots $S_{\text{pos}}(t) = s_{\text{max}} \cdot N(t)$ where s_{max} is the colonial charter's maximum administrative ratio (e.g., 0.05 for 1 elite per 20 colonists).

When $E(t) > S_{\text{pos}}(t)$: frustrated elites are politically dangerous (high EMP).

Relative elite income (Turchin Eq. 5)

$$\epsilon(t) = (1 - w(t)\lambda) / e(t)$$

Symbol	Meaning	Units
λ	labor force participation ≈ 0.55	dimensionless

When $e(t)$ is large, total elite income is diluted across more recipients → each elite gets less → ϵ falls → ϵ^{-1} rises → EMP rises. The squeeze feeds political restlessness even when the colony is nominally prosperous.

VII — Political Stress Indicator (Turchin PSI adapted)

MMP — Mass Mobilization Potential

$$\text{MMP}(t) = w(t)^{-1} \cdot f_{\text{set}}(t) \cdot A_{\text{young}}(t)$$

Symbol	Meaning	Units
$f_{\text{set}}(t) = N_{\text{set}}/N$	fraction of population in permanent settlements vs. dispersed field teams	dimensionless
$A_{\text{young}}(t)$	fraction of colony population aged 20–29	dimensionless

Rationale for f_{set} : Turchin uses urbanization N_{urb}/N as the proxy for mobilization density. In a young colony the analog is how concentrated the population is in the central settlement; dispersed field teams are hard to organize for collective action.

EMP — Elite Mobilization Potential

$$\text{EMP}(t) = \epsilon(t)^{-1} \cdot (E(t) / S_{\text{pos}}(t))$$

When elites are overproduced relative to available positions ($E > S_{\text{pos}}$), both terms amplify each other: income is diluted *and* there are more frustrated aspirants per slot.

SFD — State Fiscal Distress

$$\text{SFD}(t) = (Y(t) / G(t)) \cdot D(t)$$

$$D(t) = 1 - CL(t)$$

Symbol	Meaning	Units
Y(t)	cumulative debt owed to Earth (or charter company)	cr
G(t)	annual GDP equivalent	cr/year
Y/G	debt-to-annual-output ratio	dimensionless
D(t)	public distrust = inverse of Citizen Loyalty	dimensionless $\in [0, 1]$

A colony that is both heavily indebted to Earth *and* has low loyalty to the colonial administration scores high on SFD regardless of its nominal economic size.

Full PSI

$$\Psi(t) = w(t)^{-1} \cdot f_{\text{set}}(t) \cdot A_{\text{young}}(t) \cdot \varepsilon(t)^{-1} \cdot (E(t)/S_{\text{pos}}(t)) \cdot (Y(t)/G(t)) \cdot D(t)$$

This is the direct colony analog of Turchin's Eq. 14, with:

- w^{-1} from composite satisfaction §III
- f_{set} replacing N_{urb}/N
- A_{young} from census data
- ε^{-1} , E/S_{pos} from elite dynamics §VI
- Y/G from fiscal equation §VIII
- $D(t)$ from Citizen Loyalty §IX

VIII — Fiscal Dynamics

Revenue (GURPS RM)

$$R_{\text{rev}}(t) = w_{\text{col}}(t) \cdot N(t) \cdot RF \cdot 0.6$$

Symbol	Meaning	Units
w_{col}	average monthly credit-income per colonist	cr/person/month
RF	Revenue Factor (function of CR, from GURPS RM table)	dimensionless

RF values: CR 2 $\rightarrow$ 0.05; CR 3 $\rightarrow$ 0.10; CR 4 $\rightarrow$ 0.20; CR 5 $\rightarrow$ 0.30; CR 6 $\rightarrow$ 0.40.

$w_{\text{col}} \approx Q_I(t) / N(t)$ + shadow value of rations received ($\approx p_A \cdot r_{\text{adequate}}$).

Capital maintenance cost (WTH)

$$C_{\text{maint}}(t) = 0.00417 \cdot [AC(t) \cdot v_{AC} + IC(t) \cdot v_{IC} + MC(t) \cdot v_{MC}] \cdot \Theta(t - t_{\text{deploy}} - 120)$$

where v_x is the credit value per capital unit, $0.00417 \approx 5\%/\text{year} \div 12$, and Θ is the Heaviside step function (maintenance only starts in month 120 = year 10 of colony operation, per WTH rules).

Government operating expenditure

$$C_{\text{gov}}(t) = c_e \cdot E(t) + c_{\text{admin}} \cdot N(t)$$

Administrative costs scale with both elite headcount and total population (basic services).

Debt dynamics

$$\dot{Y}(t) = [C_{\text{gov}}(t) + C_{\text{maint}}(t)] - R_{\text{rev}}(t) - T_{\text{rev}}(t) + I_{\text{Earth}}(t)$$

Symbol	Meaning	Units
T_rev(t)	trade revenue received from Earth (lagged by $\Delta = 24$ months round-trip)	cr/month
I_Earth(t)	Earth subsidies/investment inflow	cr/month

I_Earth is policy-determined and subject to Earth-side political decisions that the colony cannot influence on timescales shorter than 12 months.

IX — Off-world Trade and Supply-Chain Lag

Export tonnage (WTH formula)

$$X(t) = Q_A_sur(t) / 100 + Q_M_sur(t) / 20 \text{ [displacement-tonnes/month]}$$

Symbol	Meaning	Units
$Q_A_sur = \max(0, Q_A - N \cdot r_min)$	surplus rations above subsistence	rations/month
Q_M_sur	materials surplus after internal construction use	tonnes/month

Trade revenue with transit lag

$$T_rev(t) = p_exp \cdot X(t - 24)$$

The round-trip lag of 24 months (12 out, 12 back for payment) means the colony's current export decisions do not yield revenue until 2 years later. This lag couples the trade loop to fiscal dynamics in a way that can produce oscillations: a production surplus today builds debt for 2 years before the credits arrive.

Import constraint: the colony can only import what fits in the return voyage. Import capacity C_{imp} [displacement-tonnes/convoy] is a hard upper bound on capital imports per cycle.

X — Citizen Loyalty and Governance Dynamics (GURPS RM)

Political Track update (WTH monthly roll)

At each monthly step, draw:

$$P_{roll}(t) \sim \text{DiscreteUniform}(1, 20) + PT(t) + MS_{adj}$$

Symbol	Meaning
$PT(t)$	current Political Track level $\in \{-3, \dots, +3\}$
MS_{adj}	administrator Management Skill modifier

Outcome:

- $P_{roll} > 15 \rightarrow PT \rightarrow PT + 1$ (capped at +3)
- $P_{roll} < 5 \rightarrow PT \rightarrow PT - 1$ (floored at -3)
- $5 \leq P_{roll} \leq 15 \rightarrow$ no change

Citizen Loyalty dynamics

$$CL(t+1) = \text{clamp}(CL(t) + \alpha_{PT} \cdot \Delta PT(t) - \alpha_{\Psi} \cdot \Psi(t) \cdot \Delta t + \alpha_{dis} \cdot \Delta CL_{dis}(t), 0, 1)$$

Symbol	Meaning
α_{PT}	loyalty gain per Political Track improvement step
α_{Ψ}	loyalty erosion per unit PSI per month

Symbol	Meaning
α_{dis}	loyalty shock from disruptions (negative)

Distrust feeds back into PSI through $D(t) = 1 - CL(t)$, creating a self-reinforcing loop: high Ψ erodes loyalty $\rightarrow D$ rises $\rightarrow$ SFD rises $\rightarrow \Psi$ rises further. This is the colonial analog of Turchin's instability spiral.

GURPS RM Realm Value

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$$V(t) = w_{\text{col}}(t) \cdot N(t) \cdot 0.6$$

Realm Value is not a direct driver in the ODE system; it serves as a summary statistic representing the colony's strategic worth to Earth (which determines I_{Earth} and immigration willingness).

XI — Disruptions (GURPS RM / Stochastic Shocks)

Disruption probability

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$$p_{\text{dis}}(t) = p_0 \cdot \Psi(t)^\kappa$$

High PSI periods are more disruption-prone. At each monthly step, if $u \sim \text{Uniform}(0,1) < p_{\text{dis}}(t)$, draw a disruption type from the GURPS RM disruption table.

Disruption effects (selected)

Disruption	Effect on state variables
Plague	$N \rightarrow N \cdot (1 - \delta_{\text{plag}})$; $CL \downarrow$ by $\delta_{\text{plag_CL}}$
Drought / Famine	$Q_A \rightarrow Q_A \cdot (1 - \delta_{\text{fam}})$ for τ_{fam} months; R_A temporarily $\downarrow$

Disruption	Effect on state variables
Corruption	$R_{\text{rev}} \rightarrow R_{\text{rev}} \cdot (1 - \delta_{\text{cor}})$; $Y \uparrow$ by diversion amount
Political Schism	Ψ spike (add δ_{sch} to Ψ for τ_{sch} months); $PT - 2$
Trade Embargo	$T_{\text{rev}} = 0$ for τ_{emb} months; $I_{\text{Earth}} = 0$ if supply convoy halted
Troublesome Flora/Fauna	$K_A \downarrow$ (carrying capacity reduced until extirpation campaign)
Pollution	$h(\text{Hab})$ temporarily $\downarrow$; N growth rate suppressed

Response-time modifier (GURPS RM)

Disruption effects decay at rate governed by the administrator's Management Skill MS and Technology Level TL:

$$\tau_{\text{resp}} = \tau_0 / (1 + MS/2)$$

A disruption shock X_{dis} decays as:

$$X_{\text{dis}}(t) = X_{\text{dis}}(0) \cdot \exp(-t / \tau_{\text{resp}})$$

Higher MS $\rightarrow$ shorter response time $\rightarrow$ disruptions clear faster.

XII — Capital Investment Dynamics

Capital purchase constraint

Capital is purchased with industrial credits and imported from Earth:

$$\Delta C_x(t) = \min(\text{credits_available}, \text{import_capacity_allocation}) / v_x$$

Capital units purchased at time t arrive $\Delta = 12$ months later (one-way transit). Newly deployed capital begins operation in month $t + 12$ and begins

incurring maintenance in month $t + 12 + 120$.

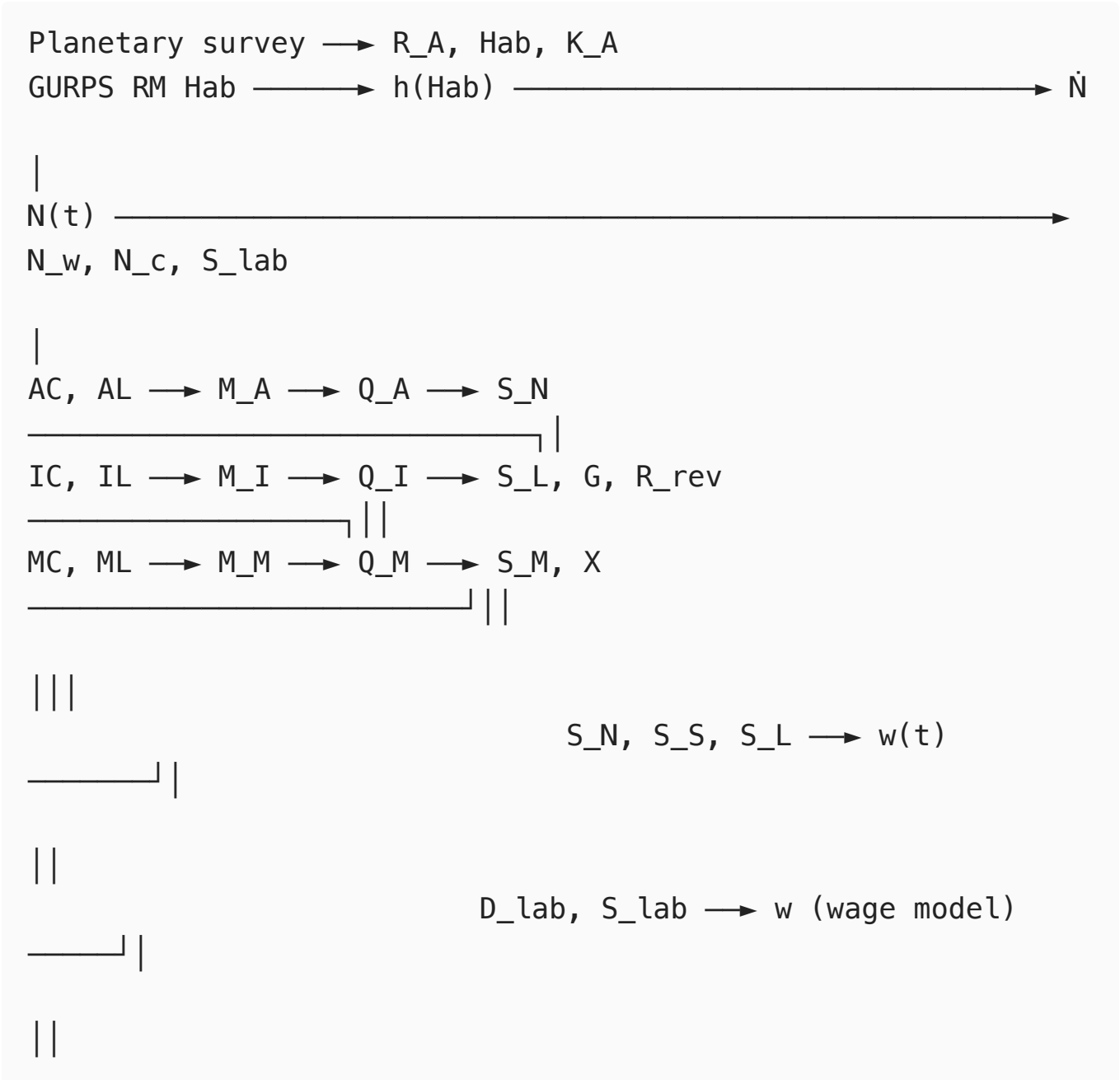
Capital depreciation

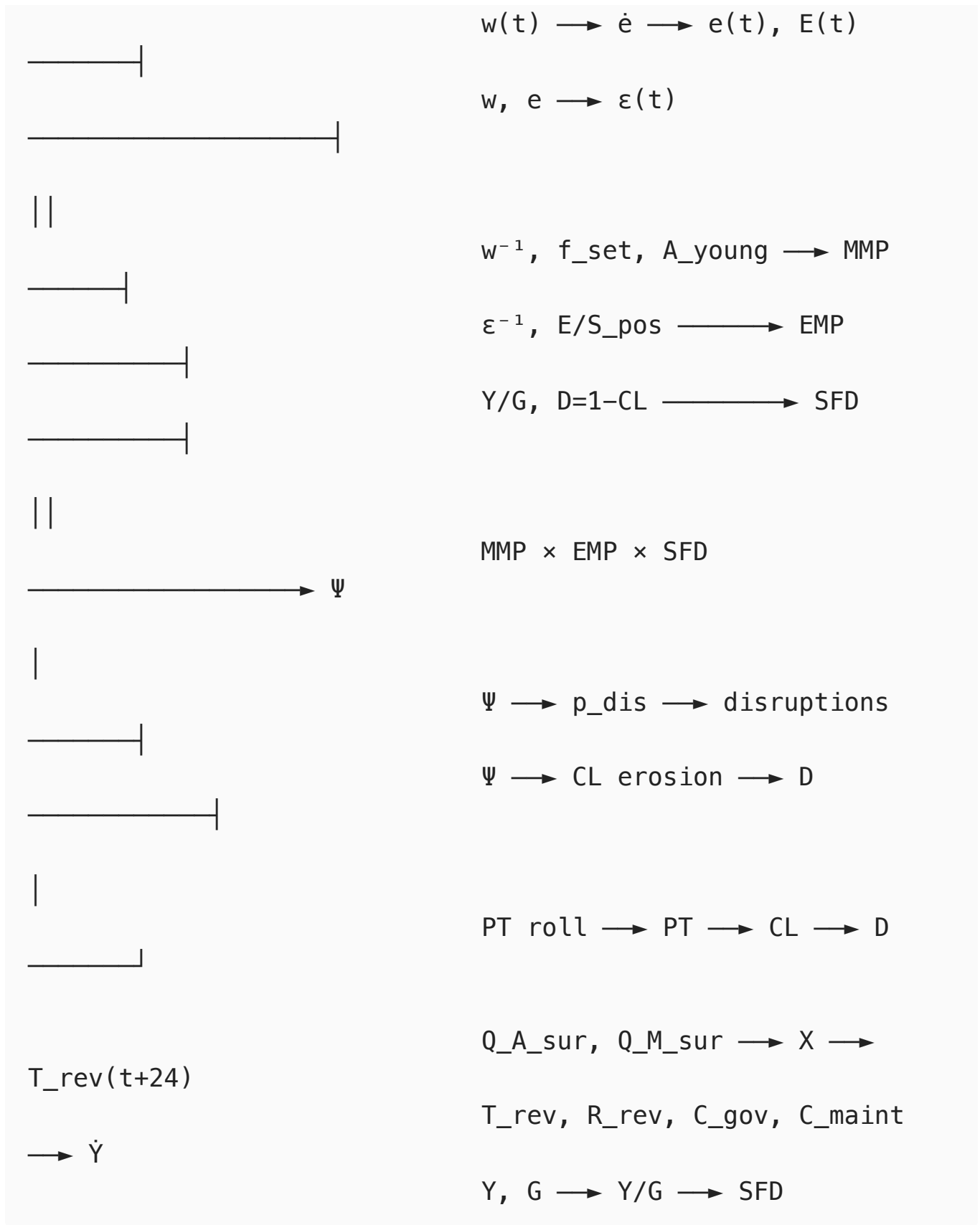
$$\dot{C}_x(t) = -\delta_{cap} \cdot C_x(t)$$

(ongoing wear beyond maintenance replacement)

If maintenance is not funded (credits insufficient), depreciation accelerates.

XIII — Complete Dependency Structure





Primary feedback loops:

1. **Low-welfare spiral:** $w \downarrow \rightarrow \dot{e} \uparrow \rightarrow e \uparrow \rightarrow \varepsilon \downarrow \rightarrow \varepsilon^{-1} \uparrow \rightarrow \text{EMP} \uparrow \rightarrow \Psi \uparrow \rightarrow \text{CL} \downarrow \rightarrow D \uparrow \rightarrow \text{SFD} \uparrow \rightarrow \Psi \uparrow \uparrow$

2. **Fiscal trap:** $Y \uparrow \rightarrow Y/G \uparrow \rightarrow SFD \uparrow \rightarrow \Psi \uparrow \rightarrow$ disruptions more frequent
 $\rightarrow$ production $\downarrow \rightarrow G \downarrow \rightarrow Y/G \uparrow\uparrow$
3. **Elite saturation:** $w \downarrow \rightarrow$ mobility $\uparrow \rightarrow E \uparrow$ past $S_{pos} \rightarrow$ frustrated elites
 $\rightarrow EMP \uparrow \rightarrow \Psi \uparrow \rightarrow$ further welfare pressure
4. **Trade-lag oscillator:** surplus production $t \rightarrow$ export $\rightarrow T_{rev}$ arrives
 $t+24 \rightarrow$ credits finance capital $\rightarrow$ output rises $t+36 \rightarrow$ welfare improves;
 if conditions reversed in interval, lags can destabilize
5. **Stabilizing loop:** PT improvement $\rightarrow CL \uparrow \rightarrow D \downarrow \rightarrow SFD \downarrow \rightarrow \Psi \downarrow$
 (requires MS-adjusted favorable rolls)

XIV — Measurable Outputs

These are the observable quantities a colonial survey or Earth oversight body could directly measure at any time step.

Output	Formula	Units	What it signals
N(t)	total population	persons	Colony size and growth trajectory
G(t)	$p_A \cdot Q_A + Q_I + p_M \cdot Q_M \times 12$	cr/year	Economic output; fiscal baseline
$\Psi(t)$	$MMP \times EMP \times SFD$	dimensionless	Integrated instability risk (leading indicator)
S_N(t)	$Q_A / (N \cdot r_{min})$	dimensionless	Nutrition adequacy; famine risk
S_S(t)	H / N	dimensionless	Shelter adequacy; overcrowding
S_L(t)	$Q_I / (N \cdot I_{min})$	dimensionless	Living standards; consumer goods availability
w(t)	$(S_N + S_S + S_L) / 3$	dimensionless	Composite welfare; mobility threshold

Output	Formula	Units	What it signals
			proximity
e(t)	E / N	dimensionless	Elite fraction; tracks overproduction
$\epsilon(t)$	$(1 - w\lambda) / e$	dimensionless	Elite income dilution
CL(t)	citizen loyalty	dimensionless	Political stability
Y(t)/G(t)	cumulative debt / annual GDP	years	Fiscal sustainability
X(t)	$Q_A_sur/100 + Q_M_sur/20$	displacement-tonnes/month	Export capacity; trade viability
T_rev(t)	$p_exp \cdot X(t-24)$	cr/month	Lagged trade revenue actually received
p_dis(t)	$p_0 \cdot \Psi^k$	probability/month	Disruption risk; useful for scenario planning
τ_resp	$\tau_0 / (1 + MS/2)$	months	Governance response speed to shocks
V(t)	$w_col \cdot N \cdot 0.6$	cr	Realm value; Earth's strategic valuation of colony

Threshold values of note

Condition	Threshold	Interpretation
Adequate nutrition	$S_N \geq 1$	No starvation pressure on growth or loyalty
Elite saturation	$e(t) > s_max$	Every elite excess beyond this is "frustrated elite"
Welfare neutral	$w(t) = w_0 = 1$	Mobility halts; system near equilibrium

Condition	Threshold	Interpretation
Fiscal danger	$Y/G > 3$	Debt exceeds 3 years of output; Earth likely to intervene
High PSI	$\Psi > \Psi_{\text{crit}}$	Colony-specific; political crisis probable within ~5 years
Loyalty collapse	$CL < 0.25$	Risk of open revolt or secession movement

XV — Parameter Summary

Parameter	Baseline value	Source	Notes
r_0	0.02 y^{-1} (= $1/600 \text{ mo}^{-1}$)	WTH	Annual population growth
λ	0.55	assumption	Working-age fraction
s_{max}	0.05	GURPS RM	Max elite fraction under standard charter
μ_0	0.005 mo^{-1}	Turchin calibrated, rescaled	Mobility parameter
w_0	1.0	Turchin	Neutral welfare threshold
β	0.5	Turchin Eq. 9	Wage labor-market elasticity
τ	3–5 mo	Turchin	Wage-stickiness lag
Δ	12 mo	setting	One-way Earth transit time
κ_N	2	assumption	Nutrition stress nonlinearity

Parameter	Baseline value	Source	Notes
p_0	0.02 / mo	GURPS RM	Base disruption probability at $\Psi = 1$
κ	1.5	assumption	PSI-disruption sensitivity exponent
τ_0	6 mo	GURPS RM RTM	Base response time at MS 0
$\varphi_{\min}$	0.5	WTH	Minimum growing-season modifier
maintenance rate	0.00417 / mo	WTH	5% capital value per year, starting month 120

Sources

- **WTH** = TNE 0311, *World Tamer's Handbook*, ch. 4 (Colonial Economics), ch. 5 (Monthly Turn)
- **GURPS RM** = *GURPS Realm Management*, ch. 1–2 (Realm Value, Revenue, Disruptions, RTM, Habitability, Education Rating)
- **3turchin** = Turchin 2013, "Modeling Social Pressures Toward Political Instability", *Cliodynamics* 4.2: 241–280; Eqs. 1–14 fully inventoried in `turchin-equations.md`